

Just like how traditionally in security, the human element is the most vulnerable aspect in terms of a system's defensive capabilities, human-AI workflow requires a more thorough and detailed analysis compared to traditionally standalone AI models. This is because the potential angles for adversarial attacks, the attack surface, just increase dramatically with the introduction of the human element. You can see this by just briefly thinking about it and numerous scenarios would pop up, ranging from intentionally manipulating the prompts to mis-aligning the features,... One of the more extreme paradigms even treats the fact that if an attacker has accessed the chat window of an LLM then that system is already breached in terms of security, regardless of prompt sanitization. This brings into mind that this is a rapidly developing field that, like the A.I. Robustness papers have demonstrated, has barely touched the surface of the human element. Only by combining both the human-alignment and the model's robustness can we truly develop a robust system in HIS-LLM.

In terms of other aspects of R3E, I would say it's more useful to draw from real world examples. We are currently in a fever of demand for AI agents with companies rushing to build agentic workflows that simulate a hybrid human-AI environment. However, if you look at the AI agents commonly seen in use, are they really elastic? Reliable? Resilience? Even with fine tuning foundational models or using RAG agents to better handle context-heavy scenarios, the user still needs to pay attention to the prompts they are giving to the agent to get the best results. In that sense, within a HIS-LLM framework, I would argue that the human element is even more important than the AI models. This is reinforced by how well smaller models perform in specialized contexts. To me, proper research regarding a new standard way of approaching hybrid workflows is vital to ensure R3E compliance in HIS-LLM.